

The aroma volatiles present in honey enhance sweetness, bringing its sweetness intensity to parity with sucrose

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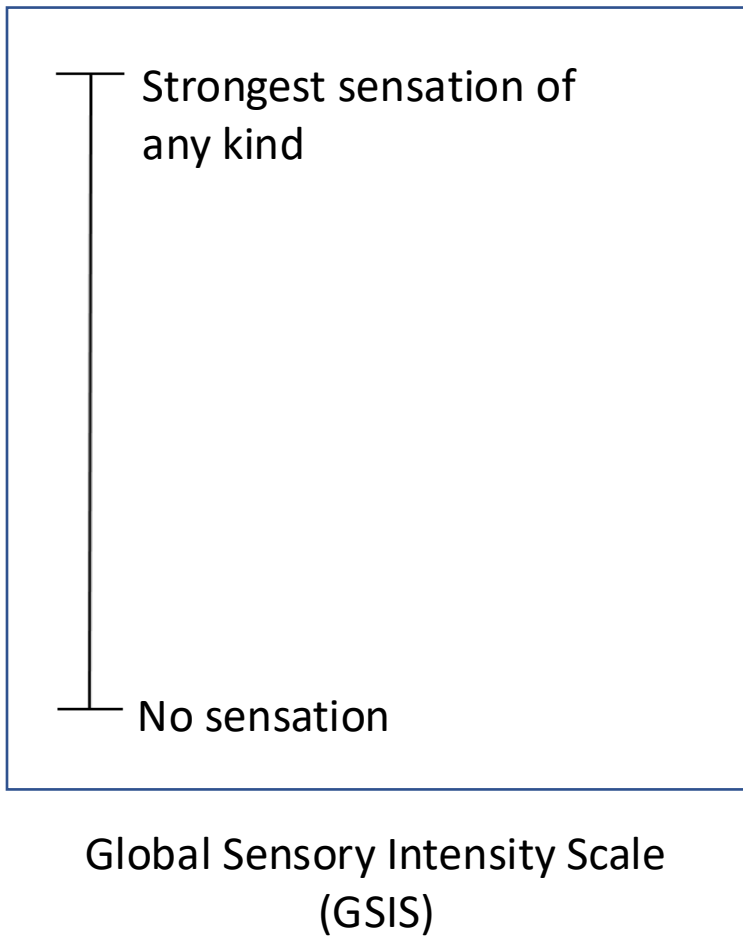


INTRODUCTION

Honey is a chemically complex natural sweetener whose sweetness has yet to be quantified. In addition to sugars, honey contains water, proteins, vitamins, minerals and volatile compounds^{1,2}. The volatiles within honey are of particular interest in the quantification of honey sweetness, as studies have shown that sweetness is not only dictated by tastants but also aroma^{3,4}. This study aims to quantify the sweetness of honey in water and food mediums and determine the impact of honey’s aroma on its perceived sweetness.

SENSORY TESTING

SEMI-TRAINED PANELISTS RATED SWEETNESS INTENSITY OF SUGAR AND HONEY SOLUTIONS



Session 1: Scale Training
n=66



1 set of Orange Blossoms solutions

Sessions 2-4: rate sets twice **with** aroma



Sessions 5-7: rate sets twice **without** aroma (nose clip worn)



n=55

5 Sweeteners: Sucrose, Alfalfa Blossom Honey, Clover Honey, Orange Blossom Honey, Wildflower Blossom Honey
Concentration Set: 12.5g/L, 25g/L, 50g/L, 75g/L, 100g/L, 125g/L



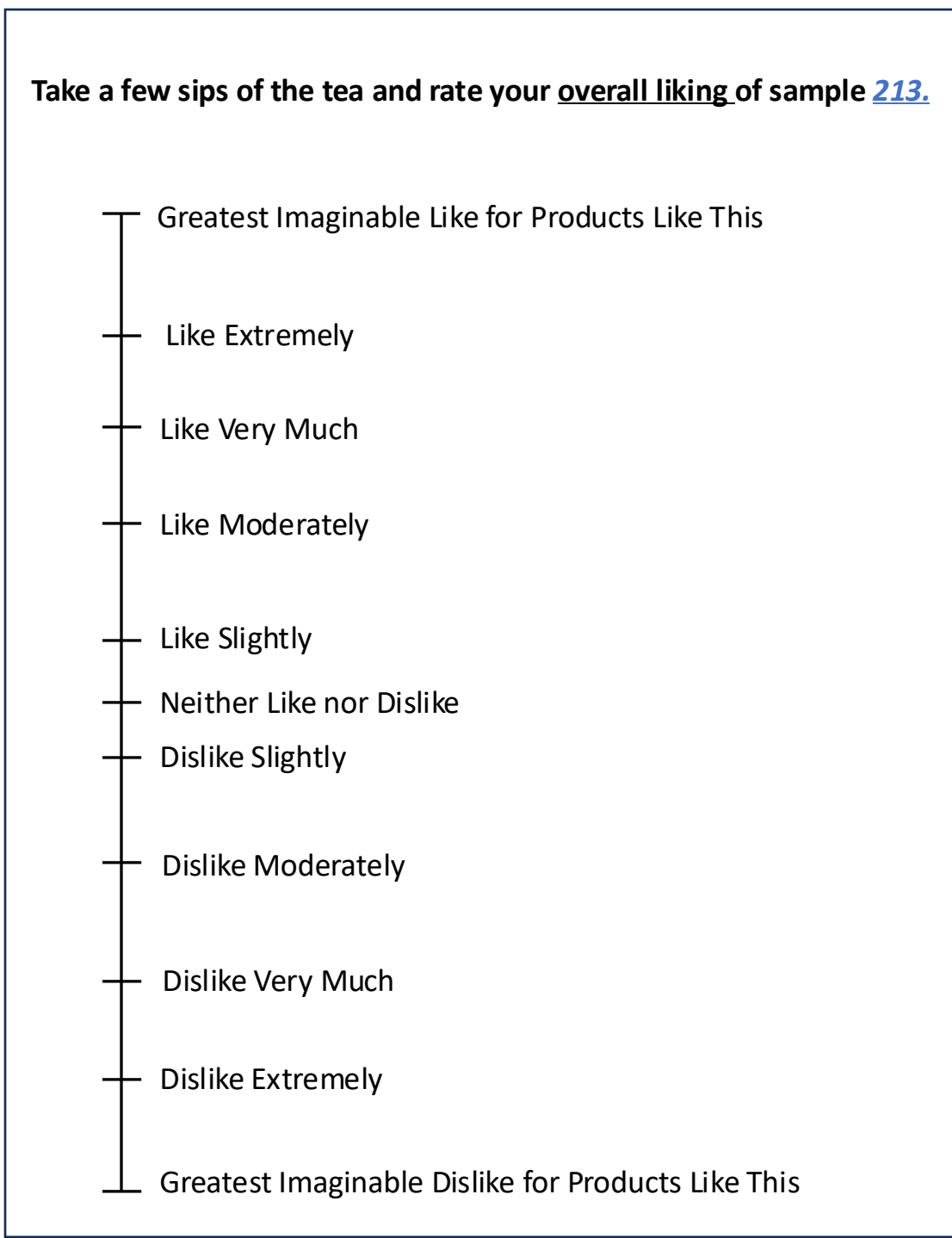
Testing room set up



Panelists rating remembered sensations during session 1.

CONSUMERS RATED OVERALL LIKING AND SWEETNESS INTENSITY OF TEA SWEETNED WITH HONEY OR SUGAR

Sugar Control			Honey			
g	kcal	tsp	Unit of equivalence	g	kcal	tsp
5.73	22.1	1.36	Mass	5.73	17.4	0.79
			Calories	7.72	22.1	1.00
			Volume	9.86	30.0	1.36



Session 1: Sugar control, equal mass honey
n=104

Session 2: Sugar control, equal volume honey, equal calories honey
n=79



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RESULTS

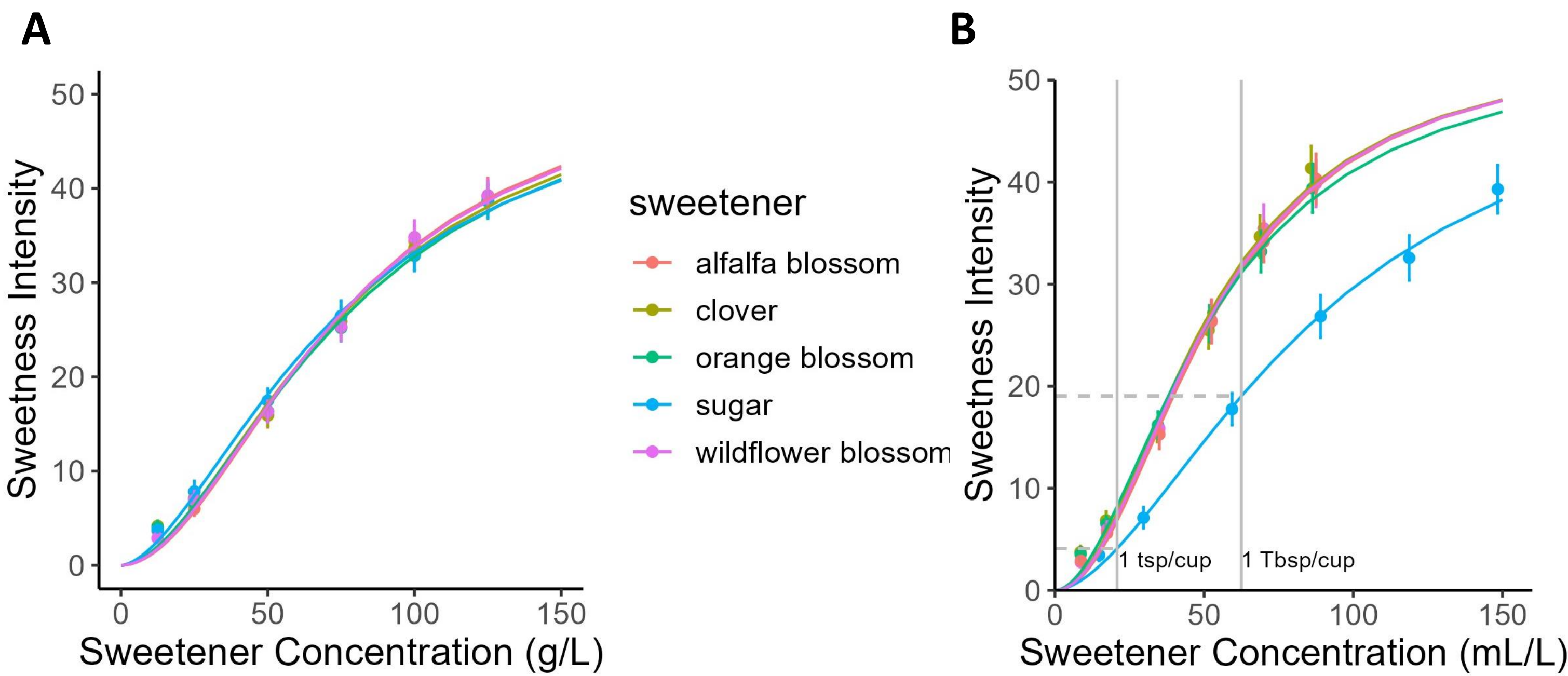


Figure 1: Mean sweetness intensity ratings with aroma as a function of sweetener concentration (A) by mass (g/L) and (B) by volume (mL/L) for all 5 sweeteners (Alfalfa Blossom Honey, Clover Honey, Orange Blossom Honey, Sucrose, and Wildflower Blossom Honey). Error bars represent the standard error of the mean (± 1 se).

When measured in units of mass, sugar and all 4 honeys are iso-sweet in water across the concentration range measured. However, when converted to units of volume, honey is sweeter than sugar in water.

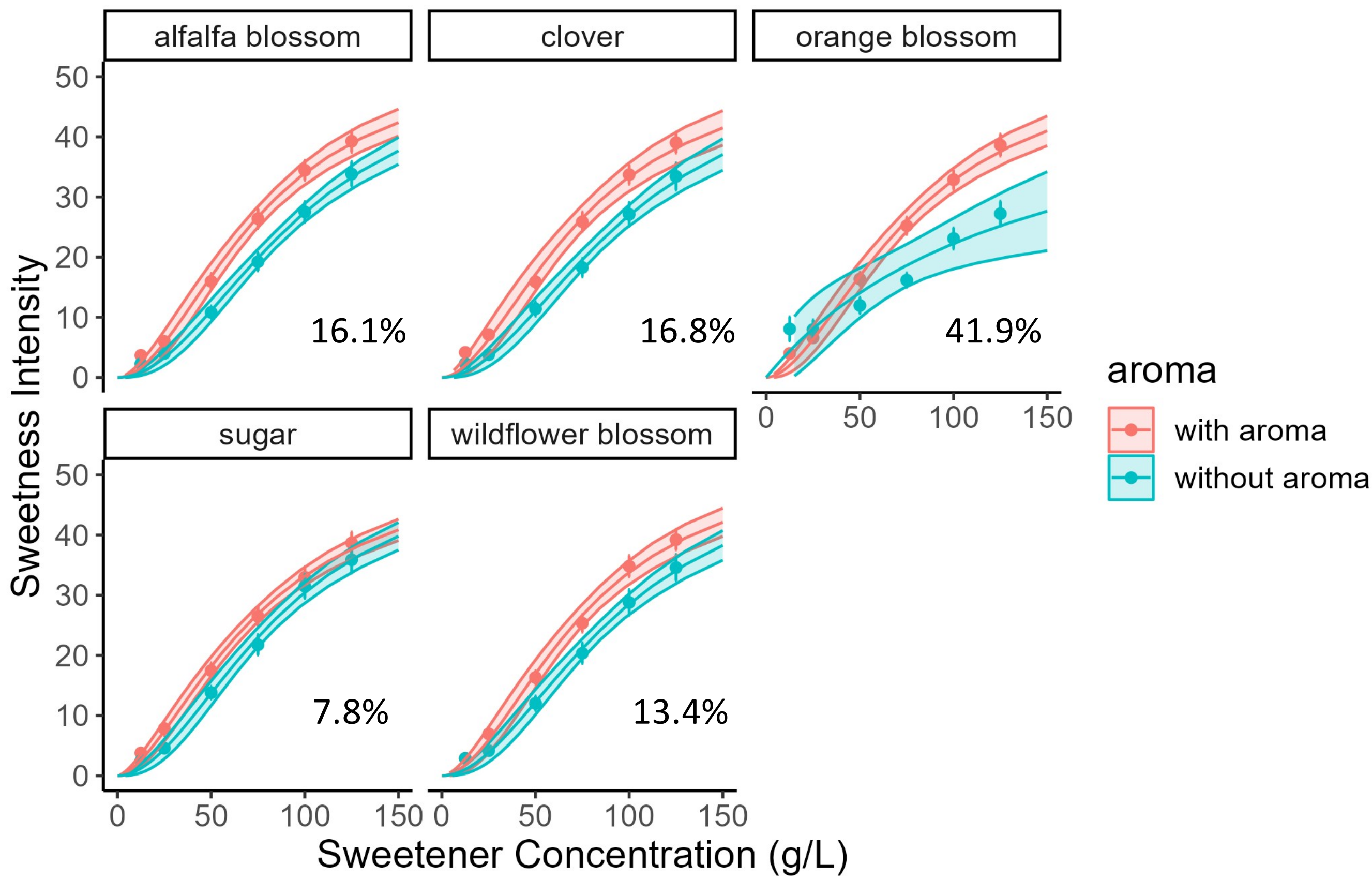


Figure 2: Mean sweetness intensity ratings as a function of sweetener concentration in grams per liter with and without aroma for each of the sweeteners: Alfalfa Blossom Honey, Clover Honey, Orange Blossom Honey, Sucrose, and Wildflower Blossom Honey. Error bars represent the standard error of the mean (± 1 se). Percent increase of perceived sweetness when aroma is present is displayed for each sweetener at a concentration of 125g/L.

There is a significant decrease in perceived sweetness intensity ($p < 1e-5$) for all sweeteners without aroma. The effect of aromas on sweetness perception is largest in the Orange Blossoms variety of honey. Without aroma, orange blossoms honey was the least sweet, however its aroma increases sweetness by up to 55.9%, bringing its sweetness to parity with the other samples.

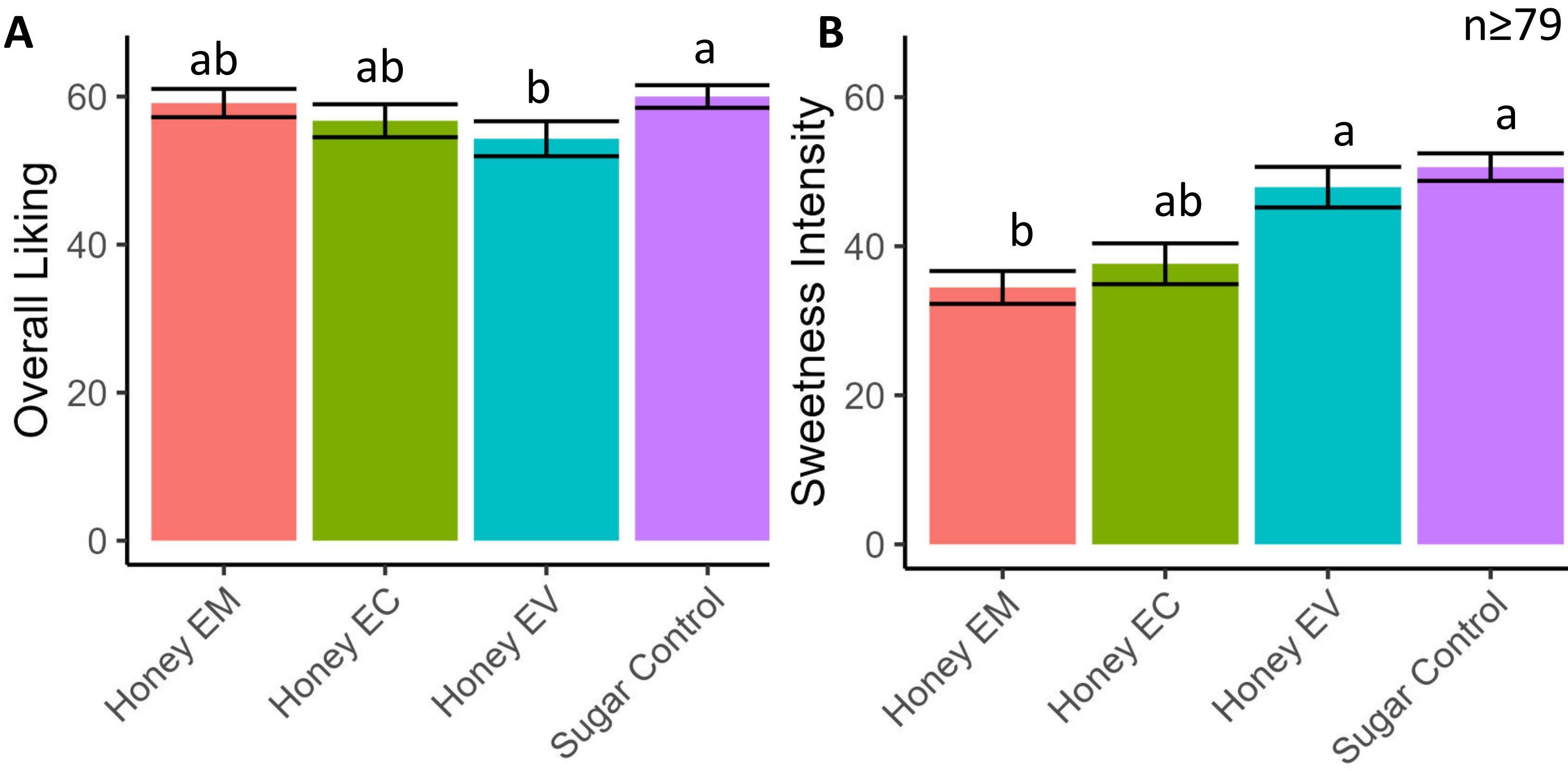


Figure 5: Mean (A) overall liking and (B) sweetness intensity ratings of black tea sweetened with sugar (Sugar control) or the unit equivalent of clover honey in: mass (Honey EM), volume (Honey EV), and kcal (Honey EC). Fisher's LSD method used to determine post hoc means separation groups. Error bars represent the standard error of the mean (± 1 se).

The sugar control had the highest mean overall liking and mean sweetness intensity, compared to all 3 honey conditions. Only the condition sweetened with an equivalent volume of honey was liked less than the control (A). Only the condition dosed with an equivalent mass of honey was less sweet than the control (B). All other honey dosed samples were at parity with the control.

CONCLUSION

Honey and sugar are equivalently sweet by mass in water. Additionally, the aroma in honey increased its perceived sweetness. In a tea medium, it is likely that the interaction of other volatiles and tastants resulted in honey’s aroma having a less potent impact on perceived sweetness.